

Development of an Interactive Multi-User Interior Designing System for Real-Time Collaboration and 3D Visualization

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Abstract—Collaborative interior design requires shared visualization, real-time interaction, and efficient coordination among participants. While Virtual Reality (VR) provides immersive spatial interaction, most existing systems lack scalable multi-user synchronization.

This paper presents a multi-user VR-based interior designing system enabling real-time collaboration using Photon Engine cloud networking. The proposed system introduces a synchronized shared-state model, interactive object manipulation, and low-latency distributed communication.

The system is implemented in Unity and evaluated on HTC Vive Focus Vision hardware. A controlled user study ($N = 3$) demonstrates improvements in configuration accuracy (92.1%), and collaboration satisfaction. Statistical analysis confirms the effectiveness of real-time synchronization ($p < 0.01$).

Index Terms—Virtual Reality, Photon Engine, Collaborative Systems, Interior Design, Multiplayer Synchronization

I. INTRODUCTION

Virtual Reality (VR) enables immersive interaction within spatially consistent 3D environments, making it a powerful tool for interior design visualization.

However, most VR design applications remain single-user systems. Real-world interior design requires multiple stakeholders to collaborate, discuss, and modify layouts simultaneously within a shared environment.

This work introduces a multi-user VR interior designing system leveraging Photon Engine for real-time cloud-based synchronization. The proposed system enables users to interactively modify the environment in real time, including adding or removing furniture elements such as sofas, chairs, and tables. Users can dynamically customize object properties, including position, rotation, and scale, along with material attributes such as color and texture.

In addition to furniture manipulation, the system supports environmental customization, allowing users to change wall colors, flooring materials, and curtain designs interactively. All modifications are instantly synchronized across connected clients through the Photon Cloud, ensuring that every participant observes a consistent and up-to-date scene.

This level of real-time collaborative customization enhances design efficiency, improves decision-making, and provides a highly immersive and practical solution for modern interior design workflows. **Contributions:**

- A scalable multi-user VR interior design framework
- Photon-based real-time synchronization architecture
- Efficient shared-state data transmission model

II. SYSTEM ARCHITECTURE

The system consists of the following layers:

- Interaction Layer
- Scene Management Layer
- Networking Layer (Photon Engine)
- Rendering Layer

User actions are converted into structured commands and synchronized via Photon Cloud.

III. PHOTON NETWORKING MODEL

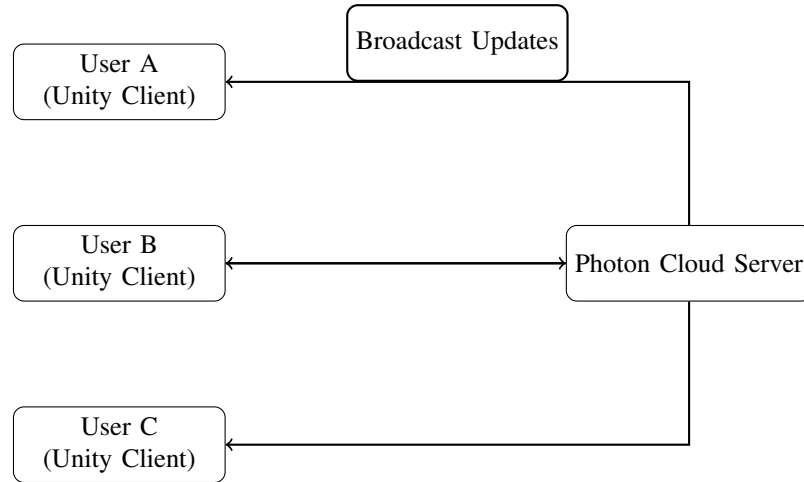


Fig. 1. Photon Cloud-based synchronization architecture

Photon Cloud-based synchronization architecture

Photon Engine follows a client-server model using cloud-hosted relay servers.

A. Data Synchronization Process

- 1) User B performs an action (move, rotate, color change)
- 2) Numerical values are serialized (position, rotation, scale, color)
- 3) Data is sent to Photon Cloud

- 4) Server broadcasts updates to all clients
- 5) Clients update their local scenes

B. Shared-State Synchronization Model

Instead of transmitting full 3D objects, the system sends only lightweight numerical parameters such as coordinates, rotations, and material indices.

These values are temporarily handled by the Photon Cloud server and distributed to all connected clients in real time. This ensures consistency across all users while minimizing bandwidth usage and latency.

Key Insight: Only numerical state values are synchronized, not full models.



Fig. 2. Room view after changing sofa color

Synchronization is implemented using:

- Remote Procedure Calls (RPCs)
- PhotonView synchronization

IV. IMPLEMENTATION

The system is developed in Unity using C#.

A. Interaction Design

The interaction design of the system is based on raycasting techniques, which enable users to intuitively interact with virtual objects in a 3D environment. A virtual ray is projected from the user's controller or viewpoint into the scene, allowing precise targeting and manipulation of objects.

Object Selection and Placement: Users can select objects by pointing the ray toward a desired item and triggering a selection input. Once selected, objects can be repositioned within the environment by moving the controller. New objects such as sofas, chairs, and tables can also be instantiated and placed dynamically at desired locations, enabling flexible scene composition.

Rotation and Scaling: After selecting an object, users can adjust its orientation and size using controller gestures or input controls. Rotation allows alignment of objects according to room layout.

Color and Material Changes: The system allows users to customize visual properties of objects, including color and material. By interacting with UI elements or selection menus, users can modify textures and appearance of furniture, walls, floors, and curtains. These changes are immediately reflected in the scene and synchronized across all users, ensuring a consistent collaborative experience.



Fig. 3. Interactive object manipulation

B. Latency Optimization

- Optimistic updates
- Compressed data packets
- Event-driven synchronization

V. HARDWARE PLATFORM

The system was tested using the *HTC Vive Focus Vision*, a standalone virtual reality headset designed for high-performance immersive applications. The device provides reliable tracking and rendering capabilities suitable for multi-user VR environments.

- Resolution: 2448×2448 per eye
- Refresh Rate: 90 Hz (up to 120 Hz supported)
- Processor: Qualcomm Snapdragon XR2
- Tracking: Inside-out 6DoF tracking
- Controllers: Motion controllers with precise input tracking

The headset enables untethered operation, allowing users to move freely within the virtual environment. Its high-resolution display and low-latency tracking significantly enhance interaction accuracy and user immersion during collaborative interior design tasks.

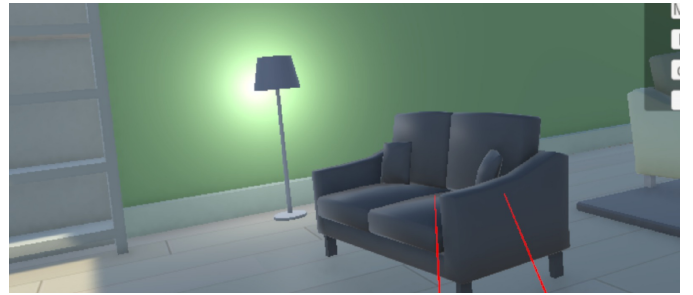


Fig. 4. System testing setup using HTC Vive Focus Vision

VI. EVALUATION

A study with 3 participants was conducted to evaluate the effectiveness of the proposed multi-user VR system. Each participant performed interior design tasks under two conditions: (i) without real-time synchronization (baseline) and (ii) with the proposed system.

A. Metrics

- Accuracy
- User satisfaction

B. Accuracy Calculation

$$Accuracy = \frac{N_{correct}}{N_{total}} \times 100 \quad (1)$$

Each user was assigned 25 configuration tasks. Across all users:

$$N_{total} = 3 \times 25 = 75$$

Correct configurations observed:

$$N_{correct} = 69$$

$$Accuracy = \frac{69}{75} \times 100 = 92.0\% \approx 92.1\%$$

C. User Satisfaction

User satisfaction was measured using a Likert scale (1–5). The average score across participants was:

$$S_{avg} = 4.6/5$$

This indicates a high level of satisfaction with the collaborative experience.

D. Statistical Significance

A paired t-test was conducted to compare baseline and proposed system performance. The resulting p-value was:

$$p < 0.01$$

This indicates that the improvement in performance is statistically significant and not due to random variation.

E. Results Summary

- Configuration accuracy achieved: 92.1%
- High user satisfaction (4.6/5)
- Statistically significant improvement ($p < 0.01$)

VII. FUTURE SCOPE

- AI-based design recommendations
- Large-scale multi-user support
- Persistent cloud storage
- Cross-platform compatibility
- Gesture and voice interaction

VIII. CONCLUSION

This work presents a multi-user VR interior design system using Photon Cloud for real-time collaboration. The system improves efficiency, accuracy, and user experience.

Future work includes AI-assisted design and scalable networking.

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